Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/09/2005
PI Name:	Dionysiou, Dionysios Ph.D.		
Project Title:	High Performance TiO2 Photocatalytic Coatings and Membranes for the Purification, Disinfection and Recycle of Water and Air in Space Applications		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	45221-0071	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/14/2003	**End Date:**	01/14/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
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Task Description:	This study deals with the application of novel methods for the fabrication of robust nanostructured TiO2 photocatalytic films for the development of efficient water and air purification systems for advanced life support in space missions and terrestrial applications. The catalytic films immobilized on a variety of ceramic, metallic, and composite substrates will be used to fabricate reactors and membrane systems that will pose unique pore orchestration for the effective capture of incident radiation, enhanced photocatalytic activity, excellent adhesive properties, good mechanical stability, controlled hydrophilicity, and anti-biological fouling capability. These immobilized films will be the crucial component for the fabrication of integrated photocatalytic-disinfection-antifoulant reactors and membranes for the room-temperature production of clean potable and hygienic water in space missions as well as for the purification of water in terrestrial applications. Alternatively, these methods will be used for the fabrication of immobilized TiO2 films aiming at the preparation of surfaces possessing self-cleaning, disinfecting, and anti-biofouling properties for air purification. These purification systems can be used as stand-alone technologies or as supplementary and complementary to existing treatment technologies. In addition, they can serve as back-up processes or as a means of emergency-respond technologies to chemical and biological terrorism. We apply a novel fabrication procedure for the preparation of high performance TiO2 films immobilized on non-porous stainless steel substrates. The procedure concerns a novel combination of the alkoxide sol-gel and the colloidal suspension (Degussa P-25) preparation routes for the synthesis of the catalyst. We optimize the P-25 concentration and calcination temperature for enhanced photocatalytic activity and good mechanical stability. The purpose of this research is to better understand the relationship of surface microstructure and film composition and their effect on the photocatalytic activity of the films. After the optimization of the modified sol-gel procedure, we evaluate the photocatalytic activity of these films for the degradation of target contaminants. Our goal is to optimize the conditions for the destruction and mineralization of these compounds with prepared TiO2 photocatalyst and to identify any inhibition factors. Another goal is the design of a new reactor with higher surface area per unit liquid volume treated, increased photocatalytic activity, and better optimization for efficient utilization of UV radiation. In addition, we apply novel nanofabrication and self-assembling procedures to prepare highly porous nanostructured photocatalytic TiO2 particles, films immobilized on nonporous glass substrates, and membranes incorporated onto porous alumina membrane supports. The objective for this task is to use self-assembling modified sol-gel methods for the fabrication of TiO2 thin films of controlled intraparticle (framework) and interparticle (textural) porosity to be used on top of non-porous or porous supports. In the latter case (i.e., porous supports), the method is used for the fabrication of nanocomposite TiO2-ceramic membranes with tailor-designed pore size (mesoporous, 1-50 nm), improved photocatalytic properties, and enhanced mechanical stability. This research is considered as an important task within our scope to develop high performance photocatalytic coatings for water purification. The completion of the tasks and objectives is crucial for the fabrication of highly efficient robust integrated advanced oxidation reactors for the purification, disinfection, and recycle of water in space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to the development of novel photocatalytic coatings and membranes for water and air treatment, purification and disinfection for Advanced Life Support in space missions, this study can lead to the development of robust photocatalytic water and air purification systems for terrestrial applications. The study can also enhance fundamental knowledge for the fabrication of novel ceramic materials, films, and membranes with anti-biofouling properties. The method for the preparation of films with super-hydrophilic properties can be applied to other support materials with desirable properties for water spreading, light transmittance, and self-cleaning properties.
Task Progress:	The project concerns various innovative approaches for the development of highly active and mechanically stable immobilized TiO2 coatings. In the second year, we continued our work on the synthesis of TiO2 catalyst with enhanced performance. First, we continue investigating the conditions for the synthesis of TiO2 films immobilized on stainless steel using the combined alkoxide sol-gel and colloidal suspension preparation routes. Following our work in the first year on the role of P-25 loading in the sol formulation, in the second year we investigated the role of calcination temperature in the range between in order to further improve the photocatalytic activity and structural properties of the coatings. Our research work included a systematic investigation of the effect of calcination temperature on the adherence, surface structure, surface chemical composition, and photocatalytic activity. Second, TiO2 coatings immobilized on stainless steel were tested for their photocatalytic activity for the destruction of organic contaminants found in Space waste streams. Specifically, the photocatalytic degradation of phenol and creatinine were studied using these immobilized TiO2 films. This study included the effect of pH and initial contaminant concentration on the degradation of the aforementioned contaminants. The degradation efficiencies for these contaminants were successful. Since NASA’s waste streams have great inherent variability and are mainly hygiene and urine wastewater, the presence of inorganic solutes is inevitable. Current research is dealing with the effect of ionic species in water on the degradation of these two contaminants. Third, we investigated the synthesis, properties and photocatalytic activity of porous TiO2 particles, films and membranes using self-assembling methods. In this work, we explore an interesting nanotechnological method to fabricate highly ordered mesoporous materials with extremely high surface area, desired pore structure and size, and catalytic activity. This method is based on self-assembling of amphiphilic surfactants that form micelles and assist the titanium alkoxide precursors to self-organized in the micellar corona. As a result, the use of self-assembling surfactants in titania sol resulted in promising TiO2 materials with better structural properties including homogeneity, controlled pore size and distribution, high water permeability, high organic retention, and enhanced catalytic properties including high surface area, active anatase phase, small crystal size, and high material crystallinity. Aiming at the scaling up of such photocatalytic film and membrane reactors, we are currently investigating the combination of the separation function of TiO2 layer with its photocatalytic properties in an effort to better incorporate defect-free TiO2 layer on the substrate, even when using a single coating layer. This sol-gel method modified by varying the type and concentration of the surfactants can be useful to prepare nanostructured anatase TiO2 particles, thin films and membranes with high photocatalytic activity and desired pore structure for environmental applications and to synthesize similar crystal nanostructures of other oxide materials for applications in catalysis and separations.
Bibliography Type:	Description: (Last Updated: 10/14/2009)
Abstracts for Journals and Proceedings	Choi, H.; Dionysiou, D.D. "Preparation of nanostructured TiO2 photocatalysts using surfactant-assisted sol-gel method for environmental applications" 9th International Conference on TiO2 Photocatalysis: Fundamentals and Applications (TiO2-9) San Diego, CA , Oct-2004

Abstracts for Journals and Proceedings	Choi, H.; Dionysiou, D.D. "Preparation of anatase nanostructured TiO2 particles using surfactant-assisted sol-gel method" Environmental Chemistry Division of the 228th American Chemical Society National Meeting (ACS) Philadelphia, PA , Aug-2004
Abstracts for Journals and Proceedings	Choi, H.; Yoo, K.; Dionysiou, D.D. "TiO2 nanoparticles and films for the fabrication of photocatalytic membranes using ionic liquid- and surfactant-based self assembling sol-gel method" 8th International Conference on Inorganic Membranes (ICIM) Cincinnati, OH , Jul-2004
Abstracts for Journals and Proceedings	Antoniou, M.G.; Chen, Y.; Dionysiou, D.D. "Advanced water purification: Towards meeting NASA's requirements" 9th International conference on TiO2 Photocatalysis: Fundamentals and Applications (TiO2-9) San Diego, CA , Oct-2004
Awards	Chen,Y. "Ohio American Water Works Association (OAWWA) Research Paper Award " Sep-2004
Awards	Antoniou, M.G. "Poster competition award of the Oesper poster session of American Chemical Society-Cincinnati Chapter" Oct-2004
Presentation	Yoo, K.; Choi, H.; Chen, Y.; Dionysiou, D.D. "Environmental implications of TiO2 nanostructured materials prepared by sol-gel methods and other wet chemistry procedures" Symposium Nanotechnology and the Environment: Nano-Catalysis for Environmental Technologies, Division of Industrial and Engineering Chemistry, 227th American Chemical Society National Meeting (ACS) Mar-2004
Presentation	Chen, Y; Dionysiou, D.D. "Immobilization of nano-TiO2 photocatalytic films on stainless steel with enhanced photocatalytic activity and good mechanical stability for water" 9th International Conference on TiO2 Photocatalysis: Fundamentals and Applications (TiO2-9) Oct-2004
Presentation	Chen, Y.; Dionysiou, D.D. "The design and development of immobilized TiO2 photocatalytic films with enhanced photocatalytic activity and good mechanical stability for water purification" 66th Annual Ohio Section American Water Works Association Conference Sep-2004
Presentation	Chen, Y.; Dionysiou, D.D. "Immobilized TiO2 powder films on stainless steel prepared using combined alkoxide sol-gel and colloidal suspension preparation methods II: Role of calcination temperature" 228th ACS National Meeting Aug-2004
Presentation	Chen, Y.; Dionysiou, D.D. "Strategy for preparing high performance of TiO2 photocatalytic films immobilized on stainless steel substrate for water purification" 228th ACS National Meeting Aug-2004
Presentation	Chen, Y.; Dionysiou, D.D. "Development of high performance TiO2 photocatalytic coatings for the purification, disinfection, and recycle of water and air in space applications" Habitation 2004- International Conference on Space Habitation Research and Technology Development Jan-2004